

Q.NO:10

DS	Session	Searching	Question Information	Level 1	Challenge 10
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Question Description:

An array A contains integers with the following constraints:

'A' contains elements in sorted order.

Integer i occurs $i \cdot \text{floor}(\sqrt{i}) + \text{ceil}(i/2)$ times in the array.

All elements are greater than or equal to 1.

Fathima has given Q queries of type:

L R: Find the number of distinct values present in subarray $A[L \dots R]$.

Note: 1-based indexing is followed.

Constraints:

$1 \leq Q \leq 10^5$

$1 \leq L \leq R \leq 10^{13}$

Input Format:

The first line contains an integer Q denoting the number of queries.

Next Q lines contains two space-separated integers L, R , denoting the query.

Output Format:

Print the output in separate lines contains required number of distinct values.

Example

Sample Input:

2

1 3

1 6

Sample Output:

2

3

Explanation

First few elements of the array A are 1, 1, 2, 2, 2, 3, 3, 3, 3, 3, ...

- For Query 1:-
 - Number of distinct elements in subarray $A[1 \dots 3]$ is 2.
- For Query 2:-
 - Number of distinct elements in subarray $A[1 \dots 6]$ is 3.

Logical Test Cases

Test Case 1

INPUT (STDIN)

2
3 3
2 6

EXPECTED OUTPUT

1
3

Test Case 2

INPUT (STDIN)

3
3 3
2 6
5 4

EXPECTED OUTPUT

1
3
1

Code:

```
#include <stdio.h>
#include <math.h>

int main()
{
    int Q;
    scanf("%d", &Q);

    while (Q--)
    {
        long long L, R;
        scanf("%lld %lld", &L, &R);

        long long i = 1;
        long long pos = 1;
        int ans = 0;

        while (pos <= R)
        {
            long long root = (long long)sqrt(i);

            long long count = i * root + (i + 1) / 2;

            long long end = pos + count - 1;

            if (end >= L)
                ans++;

            if (pos > R)
                break;

            pos = end + 1;
            i++;
        }

        printf("%d\n", ans);
    }

    return 0;
}
```

Output:

Test_case 1

```
● aadityajaiswal@Aaditya-MacBook-Air:~$  
2  
1 3  
2  
1 6  
3
```

Test_case 2

```
● aadityajaiswal@Aaditya-MacBook-Air:~$  
3  
3 3  
1  
2 6  
3  
5 4  
1
```